

Hungry

COMPLETE FEATURE OVERVIEW

Smart Fridge AI · React + Supabase + Gemini · May 2026

Features Currently in the App

| PANTRY

- **Manual item entry** with AI name sanitization — "Tyson Chicken" → chicken
- **Receipt scanning** — photo → Gemini OCR → bulk add all items
- **Barcode text entry** + online lookup with price pull
- **Camera barcode scanner** with animated scanning line and sound on success
- **Leftover Recon** — snap a photo of a container, AI identifies the food, confirm and log it as a prepared meal
- **Voice Inventory** — speak a grocery list aloud, AI parses and bulk-adds all items
- **Category system**: Proteins, Dairy & Eggs, Fruits, Vegetables, Beverages, Snacks, Frozen, General
- **Drag-and-drop recategorization** between category buckets
- **Auto-estimated expiry dates** based on item type (online data)
- **Expiry badges** — orange pulse when an item is within 7 days of expiry
- **Price tracking** per item (pulled from barcode or entered manually)
- **Nutrition facts** per ingredient (actual from barcode, estimated from lookup table)
- **Ingredient card modal** — edit name, amount, expiry, price, category; view nutrition
- **Household vs personal** item toggle per ingredient
- **Venmo split** — request Venmo payment for household-shared items
- **Estimated inventory value** displayed in pantry header
- **Offline sync queue** — changes made offline sync to Supabase via IndexedDB when back online

| RECIPE EXPLORER

- **Spoonacular API:** 50 recipes × 16 cuisines fetched in parallel
- **TheMealDB:** A–Z letter search supplements Spoonacular
- **Category filters:** All, Breakfast, Lunch, Dinner, Snack, Dessert, Vegetarian, Vegan
- **Cuisine filters:** Indian, Chinese, Mexican, Japanese, Korean, Jamaican, Latin, African, Mediterranean
- Both filters stack independently (AND logic — e.g. Vegetarian + Indian simultaneously)
- **Pantry match percentage** shown on every recipe card
- Recipe **search bar** with debounced filtering
- **Session-storage caching** — no re-fetch on revisit within the same browser session
- All recipe and ingredient names **properly title-cased**
- **Dietary restrictions auto-filter** — setting Vegetarian, Gluten-Free, Halal, etc. in Settings silently filters the recipe browser

| RECIPE MODAL

- Full ingredient list with pantry match highlighting
- Add missing ingredients to shopping list (name only, no duplicates)
- Serving size multiplier
- Star / save recipe with yellow indicator
- Share recipe via deep-link URL
- AI ingredient substitution
- Launch cooking mode

| COOKING MODE

- Step-by-step display with progress counter
- Voice navigation: "Next", "Back", "Repeat", "Stop"
- Text-to-speech reads each step aloud
- Mic stays active across all steps — no re-press
- **Wake Lock** — screen stays on while cooking, no accidental dimming

| AI RECIPE GENERATOR ↗

- Ingredient picker sorted by expiry (soonest first)
- Select All / Clear controls
- **Recipe mode** — generates a single one-off recipe
- **Meal Prep mode** — batched weekly prep plan shown as color-coded accordions
- Respects dietary restrictions and nutrition goal from Settings

| SHOPPING LIST

- Manual add with duplicate prevention
- Check off / remove items
- **List View** (flat, default)
- **Store View** — items organized by aisle (Produce, Dairy & Eggs, Meat & Fish, Frozen, Beverages, Snacks & Bakery, Pantry)
- Monthly budget limit with inline edit

ANALYTICS DASHBOARD

- **Nutritional Overview** — protein, carbs, fat with percentage distribution bars
- **AI Nutrition Coach** — select a macro goal (Protein / Carbs / Fat / Balance) → ingredient shopping suggestions + recipe ideas with add/star actions
- **Smart Meal Prep Batches** — AI-generated weekly prep plan from pantry contents, shown as expandable accordion cards with shared ingredients and cooking tips
- **Eco-Score & Waste** — Green Chef / Eco Keeper / Getting There / Needs Work rating; expiring items list with \$ at risk; expired items with \$ lost; freshness summary grid
- **Spending Breakdown** — inventory value, projected spend, stock efficiency %, monthly budget progress bar with over-budget warning, stores shopped count

SAVED RECIPES

- Star/unstar from anywhere (yellow star when saved)
- Search saved recipes
- Filter by category and cuisine (independent)

SHOPPING SUGGESTIONS

- Near-match recipes (have most ingredients)
- Shows missing items per recipe
- Open full recipe from suggestion
- Add missing items to shopping list

HOUSEHOLD & SETTINGS

- Create / join households via invite code
- Multiple households, switch active, delete
- Per-household monthly budget
- Items can be personal or shared
- Display name (defaults to "Chef" if unset)
- Body stats: age, weight, height
- Dietary restrictions (9 options): Vegetarian, Vegan, Gluten-Free, Halal, Kosher, Dairy-Free, Nut-Free, Low-Carb, High-Protein
- Nutrition goals: Balanced, High Protein, Low Carb, Low Fat, Build Muscle, Lose Weight
- Restrictions auto-filter recipe browser + inform AI generator

AUTH & PLATFORM

- Email / password sign in and sign up
- Time-of-day personalized greeting
- Installable PWA with service worker
- Apple home screen / splash screen support
- Haptic feedback on key interactions
- Deep linking — shared recipe URLs open directly
- Local storage cache for instant offline display
- Netlify deployment + serverless function proxies (Gemini, Spoonacular)

Features Requiring a Native iOS App

Why these can't ship as a PWA

The following features depend on Apple platform APIs that are not exposed to web browsers — they require a compiled Swift/SwiftUI app distributed through the App Store. The web equivalents already built (Wake Lock, Store View, Voice Inventory, Leftover Recon) cover the user-facing need as closely as the browser allows.

FEATURE	WHAT IT WOULD DO	REQUIRES (NATIVE API)
Dynamic Island + Lock Screen Live Activity	Show current cooking step, timer, and next ingredient on the Dynamic Island and Lock Screen without opening the app	Swift <i>ActivityKit</i>
Push Notifications (closed app)	Alert "Your spinach expires tomorrow — save \$4 with this recipe" even when the app is fully closed	Apple Push Notification service (APNs)
Geofencing Store Notifications	Detect when a household member arrives at a grocery store and notify others: "Omkar is at Whole Foods — anything to add?"	<i>CoreLocation</i> background mode + <i>UNUserNotificationCenter</i>
Siri Integration	"Hey Siri, add milk to Hungry" or "What can I make for dinner?" parsed directly into pantry/recipe actions	Swift <i>SiriKit</i> App Intents
Home Screen & Lock Screen Widgets	Glanceable expiring items, today's prep plan, or shopping list count without opening the app	Swift <i>WidgetKit</i>
Apple Watch Complications	Current cooking step on the wrist, or expiring pantry items at a glance	watchOS target + <i>ClockKit</i>
HealthKit Nutrition Sync	Automatically push daily macro totals (protein, carbs, fat) to Apple Health	<i>HealthKit</i> entitlement
Face ID / Touch ID Login	Biometric authentication instead of email/password on app open	Swift <i>LocalAuthentication</i>
ARKit Fridge Scan	Point the camera at an open fridge to identify and log all visible items at once using augmented reality	<i>ARKit</i> + <i>Vision</i> framework
On-Device Food Recognition	Private, instant, offline food identification — no cloud API round-trip needed	Apple <i>CoreML</i> with custom food model